Brutal Practice Paper B19

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. An immersion rod heats 2 litres of water, raising it by 8 C every minute. The same rod is put into 5 litres of water at 26 C. If the heat it gives each minute is shared over all the water, how many minutes will it take to reach 50 C? [Heat]

- **(A)** 5 minutes
- **(B)** 7.5 minutes
- **(C)** 3 minutes
- **(D)** 15 minutes
- **(E)** 9.6 minutes
- **(F)** 8 minutes
- **(G)** 12 minutes

2. The mean of five numbers is 28. Four of them are 25, 31, 20 and 34. What is the fifth number? [Data Handling]

- **(A)** 25
- **(B)** 28
- **(C)** 30
- **(D)** 27.5
- **(E)** 30.5
- **(F)** 35
- **(G)** 110

3. At a treatment plant wastewater passes through these jobs: removing large floating objects, letting heavy sand settle, letting smaller solids settle as sludge, and using air-loving bacteria to clean it. Which order is correct? [Wastewater Story]

- **(A)** Aeration, then settling tank, then grit chamber, then bar screens
- **(B)** Grit chamber, then bar screens, then aeration, then settling tank
- **(C)** Chlorination, then bar screens, then grit chamber, then aeration
- **(D)** Settling tank, then bar screens, then grit chamber, then aeration
- **(E)** Bar screens, then settling tank, then grit chamber, then aeration
- **(F)** Bar screens, then grit chamber, then settling tank, then aeration with bacteria
- **(G)** Bar screens, then aeration, then grit chamber, then settling tank

4. A recipe uses 3/4 cup of sugar per cake. How much sugar is needed for 3 cakes, written as a mixed number? [Fractions & Decimals]

- **(A)** 3/12 cups
- **(B)** 6 cups
- **(C)** 2 cups
- **(D)** 2 1/4 cups
- **(E)** 2 3/4 cups
- **(F)** 2 1/2 cups
- **(G)** 1 3/4 cups

5. At dawn a hill town is at -4 C and by 2 p.m. it reaches 11 C. On the same day a plain warms from 9 C to 29 C. Whose air warmed by more degrees, and by how much more? [Heat]

- **(A)** Both rose equally
- **(B)** The plain, by 5 C more
- **(C)** The hill, by 15 C more
- **(D)** The hill, by 5 C more
- **(E)** The plain, by 4 C more
- **(F)** The plain, by 7 C more

6. Eight friends scored 12, 18, 9, 15, 21, 14, 18 and 11 in a quiz. What is the median score? [Data Handling]

- **(A)** 13.5
- **(B)** 14
- **(C)** 15
- **(D)** 16.25
- **(E)** 18
- **(F)** 14.5
- **(G)** 14.75

7. A town produces 2000 kilolitres of wastewater a day. After treatment, 65% is reused for gardens and the rest is released to a river. How many kilolitres are released to the river each day? [Wastewater Story]

- **(A)** 35 kilolitres
- **(B)** 700 kilolitres
- **(C)** 1400 kilolitres
- **(D)** 650 kilolitres
- **(E)** 1300 kilolitres
- **(F)** 765 kilolitres

8. Which is larger, 5/8 or 7/12, and by how much? [Fractions & Decimals]

- **(A)** 7/12 is larger, by 1/24 with no difference
- **(B)** 7/12 is larger, by 1/12
- **(C)** 5/8 is larger, by 1/4
- **(D)** 7/12 is larger, by 1/24
- **(E)** They are equal
- **(F)** 5/8 is larger, by 2/20
- **(G)** 5/8 is larger, by 1/24

9. A steel spoon and a plastic spoon are left standing in the same cup of hot tea. After one minute the steel handle feels hot but the plastic handle is still cool. Which single statement best explains this? [Heat]

- **(A)** Steel conducts heat well, so heat travels up the steel handle quickly, while plastic is a poor conductor
- **(B)** Steel makes its own heat when it is warmed
- **(C)** Steel is heavier so it holds more cold
- **(D)** Plastic reflects the heat back into the tea
- **(E)** Plastic is a better conductor, so it loses its heat to the air
- **(F)** The tea only touches the steel spoon

10. The daily number of buses arriving late at a stop over a week was 3, 5, 3, 8, 3, 6 and 2. What is the difference between the range and the mode of this data? [Data Handling]

- **(A)** 9
- **(B)** 2
- **(C)** 5
- **(D)** 8
- **(E)** 6
- **(F)** 0
- **(G)** 3

11. Which list contains ONLY contaminants usually found in domestic sewage? [Wastewater Story]

- **(A)** Only sand and gravel
- **(B)** Distilled water and salt crystals only
- **(C)** Oxygen and nitrogen gas
- **(D)** Petrol, diesel and engine coolant only
- **(E)** Human waste, food scraps, soap, oil and dirt
- **(F)** Clean rainwater, fresh air and sunlight

12. A tank is 2/3 full. Then 1/4 of the FULL tank is drained out. What fraction of the full tank is left? [Fractions & Decimals]

- **(A)** 5/6
- **(B)** 1/7
- **(C)** 1/3
- **(D)** 5/12
- **(E)** 11/12
- **(F)** 1/12
- **(G)** 8/12 as 8

13. A nurse notes a child's temperature four times: 38.4 C, 37.6 C, 39.1 C and 38.0 C. Taking 37 C as the normal body temperature, by how much did the worst reading rise above normal? [Heat]

- **(A)** 0.9 C
- **(B)** 1.1 C
- **(C)** 2.6 C
- **(D)** 3.1 C
- **(E)** 1.275 C
- **(F)** 2.1 C
- **(G)** 1.4 C

14. A bar graph shows books read this month: Asha 6, Ravi 9, Mira 4, Sam 5. If each child plans to double their count next month, how many books will the four read altogether next month? [Data Handling]

- **(A)** 12
- **(B)** 24
- **(C)** 30
- **(D)** 26
- **(E)** 52
- **(F)** 44
- **(G)** 48

15. Sludge is sent to a tank where bacteria with no air break it down, giving off biogas. If 1 tonne of sludge yields 24 cubic metres of biogas, how much biogas comes from 3.5 tonnes of sludge? [Wastewater Story]

(A) 240 cubic metres
(B) 8.4 cubic metres
(C) 75 cubic metres
(D) 96 cubic metres
(E) 72 cubic metres
(F) 84 cubic metres

16. Multiply 2/5 by 3/4 and give the answer in lowest terms. [Fractions & Decimals]

(A) 8/15
(B) 6/20 only
(C) 3/20
(D) 6/9
(E) 5/9
(F) 1/2
(G) 3/10

17. On a hot afternoon at the coast, the land heats up faster than the sea. In which direction does the breeze blow near the shore, and why? [Heat]

(A) From sea to land, because the sea is warmer than the land
(B) There is no breeze, because air only moves at night
(C) Straight upward, because hot air never moves sideways
(D) From sea to land, because cool air sinks over the land
(E) From land to sea, because the land is colder than the sea by day
(F) From sea to land, because warm air over the hotter land rises and cooler sea air moves in to take its place

18. A bag has 4 red, 6 blue and 2 green marbles, all the same size. One marble is drawn without looking. What is the chance it is NOT blue? [Data Handling]

(A) 1/4
(B) 1/3
(C) 6/1
(D) 5/6
(E) 2/3
(F) 1/2

19. In the aeration tank, air is bubbled through the water. What is the main reason for adding air? [Wastewater Story]

(A) To dissolve sand and grit faster
(B) To make the water taste better
(C) To push large objects to the surface
(D) To cool the water down quickly
(E) To add chlorine to the water
(F) To help oxygen-using bacteria grow and feed on the wastes, cleaning the water
(G) To stop all bacteria from living in the tank

20. Divide 3/4 by 2. What fraction results, in lowest terms? [Fractions & Decimals]

- **(A)** 1 1/2
- **(B)** 6/4
- **(C)** 3/4
- **(D)** 1/8
- **(E)** 2/3
- **(F)** 3/8
- **(G)** 3/2

21. Two identical bottles of water, one wrapped in black paper and one in white paper, are left in bright sun. The black one warms by 12 C and the white one by 5 C. If both started at 28 C, what is the difference in their final temperatures? [Heat]

- **(A)** 12 C
- **(B)** 17 C
- **(C)** 2.4 C
- **(D)** 5 C
- **(E)** 7 C
- **(F)** 19 C
- **(G)** 68 C

22. A class of 20 students averaged 60 marks; another class of 30 students averaged 70 marks. What is the average mark of all 50 students together? [Data Handling]

- **(A)** 130
- **(B)** 66
- **(C)** 3300
- **(D)** 63
- **(E)** 65
- **(F)** 66.6
- **(G)** 64

23. A sewer pipe carries 150 litres of wastewater every minute into an empty holding tank that can take 9000 litres. How long until the tank is full? [Wastewater Story]

- **(A)** 9 minutes
- **(B)** 600 minutes
- **(C)** 45 minutes
- **(D)** 90 minutes
- **(E)** 150 minutes
- **(F)** 1.5 minutes
- **(G)** 60 minutes

24. A ribbon is 2 1/2 m long. Pieces each 5/8 m long are cut from it. How many whole pieces can be cut? [Fractions & Decimals]

- **(A)** 5 pieces
- **(B)** 4 pieces with 1/8 m left
- **(C)** 6 pieces
- **(D)** 4 pieces
- **(E)** 3 pieces
- **(F)** 2 pieces

25. Steel railway lines are laid with small gaps between them. In summer each 12 m rail gets longer by 6 mm. A track is made of 50 such rails laid end to end. What total extra length must all the gaps together allow for, in centimetres? [Heat]

- **(A)** 60 cm
- **(B)** 300 cm
- **(C)** 3 cm
- **(D)** 0.3 cm
- **(E)** 6 cm
- **(F)** 3000 cm
- **(G)** 30 cm

26. A double bar graph shows: in test 1, Neha 14 and Om 18; in test 2, Neha 20 and Om 15. Whose marks improved more from test 1 to test 2, and by how much more? [Data Handling]

- **(A)** Neha, by 9 marks more
- **(B)** Neha, by 11 marks more
- **(C)** Om, by 3 marks more
- **(D)** Both improved equally
- **(E)** Neha, by 3 marks more
- **(F)** Om, by 9 marks more
- **(G)** Neha, by 6 marks more

27. Which everyday habit best helps keep drains and treatment plants working well? [Wastewater Story]

- **(A)** Flushing cotton and plastic wrappers down the toilet
- **(B)** Putting food scraps and grease down the drain
- **(C)** Pouring leftover paint and chemicals down the drain
- **(D)** Pouring old medicines into the toilet
- **(E)** Tipping engine oil into the kitchen sink
- **(F)** Throwing used cooking oil and tea leaves in the bin, not down the sink
- **(G)** Washing wet paint brushes in the wash basin

28. A ribbon is 0.045 metres wide. A roll is made of 100 such ribbons side by side. What is the total width in metres, and in centimetres? [Fractions & Decimals]

- **(A)** 4.5 m, which is 4.5 cm
- **(B)** 450 m, which is 45000 cm
- **(C)** 4.5 m, which is 45 cm
- **(D)** 45 m, which is 4500 cm
- **(E)** 0.045 m, which is 4.5 cm
- **(F)** 4.5 m, which is 450 cm

29. When a pan of water is heated only at the bottom, the whole pan still gets hot. Which sequence correctly describes how the heat spreads through the water? [Heat]

- **(A)** Heat rays travel straight up through the water as radiation
- **(B)** The top water heats first and warms the water below it
- **(C)** Heat passes from molecule to molecule without the water moving, as in a metal rod
- **(D)** The water near the sides heats first and moves inward
- **(E)** Hot water at the bottom rises, cooler water sinks to take its place, and this circulation carries heat through all the water
- **(F)** Bubbles carry cold downwards while heat stays at the bottom
- **(G)** Cold water rises because it is lighter than hot water

30. The mean weight of 6 boxes is 14 kg. One 14 kg box is replaced by a 26 kg box. What is the new mean weight? [Data Handling]

- **(A)** 20 kg
- **(B)** 16.5 kg
- **(C)** 16 kg
- **(D)** 15 kg
- **(E)** 12 kg
- **(F)** 26 kg

31. A village spends 8000 rupees a month running its small treatment plant. After a repair, the cost drops by 15%. What is the new monthly cost? [Wastewater Story]

- **(A)** Rs 6800
- **(B)** Rs 6850
- **(C)** Rs 1360
- **(D)** Rs 7800
- **(E)** Rs 1200
- **(F)** Rs 6000

32. From a 5 metre cloth, pieces of 1.35 m and 2.4 m are cut off. What length of cloth is left? [Fractions & Decimals]

- **(A)** 1.45 m
- **(B)** 1.35 m
- **(C)** 3.75 m
- **(D)** 0.25 m
- **(E)** 1.15 m
- **(F)** 1.25 m

33. A litre of water at 70 C is mixed with a litre at 30 C in an insulated jug. Ignoring losses, the result is the average of the two. A child then pours in a third equal litre at 20 C. What is the final temperature of all three litres together? [Heat]

- **(A)** 45 C
- **(B)** 30 C
- **(C)** 60 C
- **(D)** 50 C
- **(E)** 40 C
- **(F)** 120 C

34. A spinner has 8 equal parts numbered 1 to 8. What is the chance the arrow lands on a number that is both even and greater than 4? [Data Handling]

- **(A)** 1/2
- **(B)** 5/8
- **(C)** 3/4
- **(D)** 1/4
- **(E)** 3/8
- **(F)** 2/8 as 2

35. In places with no underground sewers, which low-cost arrangement treats human waste right where it is produced? [Wastewater Story]

- **(A)** A septic tank or a composting (vermi-processing) toilet
- **(B)** A water meter fitted on the house
- **(C)** Throwing it into a pond
- **(D)** A long pipe straight to the nearest river
- **(E)** A large bar screen only
- **(F)** A chlorination tank only

36. A 4.5 litre bottle of juice is shared equally into 6 glasses. How much juice is in each glass, in litres and in millilitres? (1 L = 1000 mL) [Fractions & Decimals]

- **(A)** 0.6 L, which is 600 mL
- **(B)** 0.8 L, which is 800 mL
- **(C)** 0.75 L, which is 75 mL
- **(D)** 0.75 L, which is 750 mL
- **(E)** 0.075 L, which is 75 mL
- **(F)** 1.5 L, which is 1500 mL

37. A laboratory thermometer is marked from -10 C to 110 C. A clinical thermometer is marked from 35 C to 42 C. How many degrees wider is the lab thermometer's range than the clinical one's? [Heat]

- **(A)** 100 C
- **(B)** 7 C
- **(C)** 120 C
- **(D)** 110 C
- **(E)** 127 C
- **(F)** 113 C
- **(G)** 75 C

38. The rainfall on four days was 2.5, 0.0, 3.1 and 1.6 cm. What was the mean daily rainfall over these four days? [Data Handling]

- **(A)** 1.7 cm
- **(B)** 2.0 cm
- **(C)** 1.8 cm
- **(D)** 1.6 cm
- **(E)** 7.2 cm
- **(F)** 0.9 cm

39. Why are bar screens placed at the very start of a treatment plant rather than later? [Wastewater Story]

- **(A)** To colour the water for testing
- **(B)** To kill germs with chlorine first
- **(C)** To turn sludge into biogas
- **(D)** To add oxygen for bacteria right away
- **(E)** To cool the incoming water
- **(F)** To catch large objects like rags and sticks before they damage pumps and later stages

40. Four runners' times (in seconds) are 12.5, 12.45, 12.504 and 12.4. Who is the FASTEST (smallest time)? [Fractions & Decimals]

- **(A)** They are all equal
- **(B)** The runner with 12.45 s
- **(C)** The runner with 12.5 s
- **(D)** The runners with 12.504 s and 12.5 s tie
- **(E)** The runners with 12.45 s and 12.5 s tie
- **(F)** The runner with 12.4 s
- **(G)** The runner with 12.504 s

41. A cup of hot milk cools from 80 C, losing 4.5 C every minute at first. If it kept cooling at that steady rate, what would its temperature be after 6 minutes? [Heat]

- **(A)** 53 C
- **(B)** 49.5 C
- **(C)** 44 C
- **(D)** 53.5 C
- **(E)** 27 C
- **(F)** 26 C

42. Night temperatures one week were 4, -2, 0, 3, -5, 1 and 2 C. What is the range of these temperatures? [Data Handling]

- **(A)** 9 C
- **(B)** -9 C
- **(C)** 6 C
- **(D)** 4 C
- **(E)** 1 C
- **(F)** 3 C

43. A water sample holds 0.8 grams of dissolved waste in every litre. After treatment this falls to 0.05 grams per litre. By how many grams per litre did the waste drop? [Wastewater Story]

- **(A)** 0.03 g per litre
- **(B)** 0.84 g per litre
- **(C)** 0.3 g per litre
- **(D)** 0.04 g per litre
- **(E)** 0.75 g per litre
- **(F)** 0.85 g per litre
- **(G)** 0.45 g per litre

44. A pen costs Rs 12.50 and a notebook Rs 28.75. You buy 3 pens and 2 notebooks and pay with a 100-rupee note. How much change do you get? [Fractions & Decimals]

- **(A)** Rs 41.25
- **(B)** Rs 5.00
- **(C)** Rs 7.50
- **(D)** Rs 4.25
- **(E)** Rs 8.75
- **(F)** Rs 5.25

45. A cook needs a pan that heats food fast but a handle that stays cool to hold. Which pair of materials is the best choice for the pan body and the handle? [Heat]

- **(A)** Wooden body with an aluminium handle
- **(B)** Copper body with an iron handle
- **(C)** Plastic body with a wooden handle
- **(D)** Glass body with a steel handle
- **(E)** Aluminium body with a wooden handle
- **(F)** Wooden body with a plastic handle

46. Six houses have 2, 3, 3, 4, 4 and 20 people. Which is larger, the mean or the median, and by how much? [Data Handling]

- **(A)** The mean, by 3.5
- **(B)** The median, by 3.5
- **(C)** The median, by 2.5
- **(D)** The mean, by 16.5
- **(E)** They are equal
- **(F)** The mean, by 2.5

47. Which statement best defines 'sewage'? [Wastewater Story]

- **(A)** Only the solid sludge left at a plant
- **(B)** The sand removed in a grit chamber
- **(C)** Wastewater released by homes, industries and other users, carrying dissolved and suspended impurities
- **(D)** Fresh river water before any use
- **(E)** Treated water that is safe to drink
- **(F)** The gas given off by digested sludge

48. A tile is 0.6 m long and 0.25 m wide. What is its area in square metres? [Fractions & Decimals]

- **(A)** 0.015 sq m
- **(B)** 0.15 sq m
- **(C)** 0.30 sq m
- **(D)** 1.5 sq m
- **(E)** 0.85 sq m
- **(F)** 15 sq m

49. It takes 3 minutes on a stove to warm 600 g of water by a certain amount. How long, at the same steady rate of heating, to warm 1.5 kg of the same water by the same amount? [Heat]

- **(A)** 4.5 minutes
- **(B)** 5 minutes
- **(C)** 6 minutes
- **(D)** 2.5 minutes
- **(E)** 9 minutes
- **(F)** 1.2 minutes
- **(G)** 7.5 minutes

50. On a bar graph each unit of height stands for 5 cars. One bar is 7 units tall and another is 4 units tall. How many more cars does the taller bar represent? [Data Handling]

- **(A)** 55 cars
- **(B)** 35 cars
- **(C)** 15 cars
- **(D)** 3 cars
- **(E)** 20 cars
- **(F)** 11 cars
- **(G)** 12 cars

51. A settling tank receives 500 litres of sewage in which 4% by volume is settleable solids. How many litres of solids settle out? [Wastewater Story]

- **(A)** 200 litres
- **(B)** 25 litres
- **(C)** 480 litres
- **(D)** 20 litres
- **(E)** 4 litres
- **(F)** 40 litres
- **(G)** 125 litres

52. How many 0.25 litre cups can be completely filled from a 6 litre container? [Fractions & Decimals]

- **(A)** 240 cups
- **(B)** 30 cups
- **(C)** 2.4 cups
- **(D)** 6 cups
- **(E)** 12 cups
- **(F)** 1.5 cups
- **(G)** 24 cups

53. Late at night near the coast the land has cooled below the temperature of the sea. Which way does the breeze blow now, and what is it called? [Heat]

- **(A)** From sea to land - a sea breeze, because the land is now warmer
- **(B)** From land to sea, because the sea has frozen over
- **(C)** From sea to land - a land breeze
- **(D)** No breeze, because the temperatures are equal at night
- **(E)** From land to sea - a land breeze, because air over the warmer sea rises and cooler land air moves out to replace it
- **(F)** From land to sea - a sea breeze
- **(G)** Straight upward over the sea, with no sideways flow

54. A drawing pin can land 'point up' or 'point down', but tests show point-down happens far more often. Are these two results equally likely, and is it fair to give each a 1/2 chance? [Data Handling]

- **(A)** Yes - chance is always 1/2 for two options
- **(B)** No - both have a chance of 1
- **(C)** No - point down is impossible
- **(D)** No - point up is certain
- **(E)** No - the two results are not equally likely, so neither has a 1/2 chance
- **(F)** Yes - every coin-like object is fair
- **(G)** Yes - with two results each must have a 1/2 chance

55. After the water has been cleared and treated by bacteria, it is often passed through chlorine or ultraviolet light before release. What is the main purpose of this last step? [Wastewater Story]

- **(A)** To let sand settle out
- **(B)** To add nutrients back to the water
- **(C)** To colour the water blue
- **(D)** To remove large sticks and rags
- **(E)** To cool the water before release
- **(F)** To kill remaining disease-causing germs before the water is released

56. Three parcels weigh 1.2 kg, 0.85 kg and 2.305 kg. What is their total weight? [Fractions & Decimals]

- **(A)** 4.255 kg
- **(B)** 4.355 kg
- **(C)** 4.345 kg
- **(D)** 4.40 kg
- **(E)** 4.355 g
- **(F)** 4.5 kg

57. A weather diary lists midday temperatures for five days as 31, 34, 29, 33 and 28 C. The 'normal' midday value for that week is taken as 30 C. By how much is the five-day average above or below normal? [Heat]

- **(A)** 1 C above normal
- **(B)** 1 C below normal
- **(C)** 4 C above normal
- **(D)** 5 C above normal
- **(E)** Exactly normal
- **(F)** 31 C above normal
- **(G)** 0.5 C above normal

58. A car travels 60 km in the first hour, 90 km in the second and 30 km in the third. What is its average speed over the three hours? [Data Handling]

- **(A)** 50 km/h
- **(B)** 45 km/h
- **(C)** 60 km/h
- **(D)** 70 km/h
- **(E)** 90 km/h
- **(F)** 55 km/h
- **(G)** 180 km/h

59. An open drain runs through a crowded area. Which chain of effects best explains why covering it and treating the sewage improves health? [Wastewater Story]

- **(A)** Open sewage is harmless, so treatment wastes money
- **(B)** Mosquitoes clean the water, so draining them is bad
- **(C)** Open drains clean themselves, so covering them spreads disease
- **(D)** Treatment adds germs to the water
- **(E)** Covering drains makes the water dirtier
- **(F)** Covered, treated sewage stops germs and mosquitoes spreading, so fewer people fall ill

60. Which of these is exactly equal to 0.375? [Fractions & Decimals]

- **(A)** 3/4
- **(B)** 3/80
- **(C)** 1/3
- **(D)** 3/8
- **(E)** 5/8
- **(F)** 1/4
- **(G)** 37.5

61. Ice taken from a freezer at -15 C is heated until it becomes water at 25 C. Through how many degrees in total does its temperature change, and at what reading does it pass the melting point of ice? [Heat]

- **(A)** A change of 40 C, passing through 100 C
- **(B)** A change of 40 C, passing through 15 C
- **(C)** A change of 60 C, passing through 0 C
- **(D)** A change of 25 C, passing through 0 C
- **(E)** A change of 10 C, passing through 0 C
- **(F)** A change of 15 C, passing through 0 C
- **(G)** A change of 40 C, passing through 0 C

62. Shoe sizes sold one day were 5, 6, 6, 7, 7 and 8. Which statement about the mode is correct? [Data Handling]

- **(A)** There are two modes, 6 and 7, as each appears most often (twice)
- **(B)** The mode is 8, the largest size
- **(C)** The mode is 6, the smallest repeated size
- **(D)** The mode is 7, the largest repeated size
- **(E)** The mode is 39, the total
- **(F)** There is no mode
- **(G)** The mode is 6.5, the average of the repeats

63. A plant collects 1.2 tonnes of wet sludge. Drying removes 75% of its mass as water. What mass of dried sludge remains? [Wastewater Story]

- **(A)** 0.3 tonnes
- **(B)** 0.075 tonnes
- **(C)** 0.25 tonnes
- **(D)** 0.45 tonnes
- **(E)** 0.9 tonnes
- **(F)** 0.6 tonnes
- **(G)** 1.2 tonnes

64. A 7-metre rope is cut into 8 equal pieces. How long is each piece, in metres? [Fractions & Decimals]

- **(A)** 0.87 m
- **(B)** 0.8 m
- **(C)** 0.875 m
- **(D)** 0.875 cm
- **(E)** 1.142 m
- **(F)** 0.78 m
- **(G)** 8.75 m

65. A vacuum (thermos) flask keeps drinks hot for hours. Its shiny silvered walls and the vacuum between them mainly stop heat loss by which means? [Heat]

- **(A)** Both the vacuum and the shine increase conduction
- **(B)** The flask cools the drink slowly by convection only
- **(C)** The vacuum stops conduction and convection, and the shiny surface cuts down radiation
- **(D)** The vacuum stops only radiation; the shine stops conduction
- **(E)** The vacuum makes heat, and the shine adds more
- **(F)** The shine stops convection while the vacuum adds radiation

66. Five workers packed an average of 48 boxes each. If four of them packed 50, 45, 52 and 40, how many did the fifth pack? [Data Handling]

- **(A)** 53
- **(B)** 240
- **(C)** 47
- **(D)** 48
- **(E)** 53.5
- **(F)** 43

67. What is the difference between a 'sewer' and 'sewerage'? [Wastewater Story]

- **(A)** A sewer is the network; sewerage is one pipe
- **(B)** A sewer is a single pipe that carries sewage; sewerage is the whole network of such pipes
- **(C)** A sewer is a tank; sewerage is a screen
- **(D)** They mean exactly the same thing
- **(E)** Sewerage is a gas; a sewer is a liquid
- **(F)** Sewerage is the treated water; a sewer is the sludge
- **(G)** A sewer carries clean water; sewerage carries dirty water

68. Petrol costs Rs 105.50 per litre. How much do 4.5 litres cost? [Fractions & Decimals]

- **(A)** Rs 110.00
- **(B)** Rs 422.00
- **(C)** Rs 474.75
- **(D)** Rs 4747.50
- **(E)** Rs 468.75
- **(F)** Rs 474.50

69. A freezer is set to -18.0 C. A power cut lets it warm up by 0.6 C every 10 minutes. After 1 hour without power, what does the freezer thermometer read? [Heat]

- **(A)** -14.4 C
- **(B)** -21.6 C
- **(C)** -14.6 C
- **(D)** -18.6 C
- **(E)** -3.6 C
- **(F)** -12.4 C
- **(G)** -15.4 C

70. Cards numbered 1 to 20 are shuffled and one is drawn at random. What is the probability of drawing a multiple of 3? [Data Handling]

- **(A)** 6/10
- **(B)** 1/2
- **(C)** 1/4
- **(D)** 3/10
- **(E)** 1/3
- **(F)** 3/20

71. A town of 5000 people produces, on average, 120 litres of wastewater per person each day. How many kilolitres of wastewater is that for the whole town daily? (1 kilolitre = 1000 litres) [Wastewater Story]

- **(A)** 600 kilolitres
- **(B)** 600000 kilolitres
- **(C)** 60 kilolitres
- **(D)** 6 kilolitres
- **(E)** 720 kilolitres
- **(F)** 6000 kilolitres
- **(G)** 500 kilolitres

72. A jump measured 3.2 m, another 3.55 m and a third 3.0 m. What is the average jump length? [Fractions & Decimals]

- **(A)** 3.375 m
- **(B)** 3.255 m
- **(C)** 3.25 m
- **(D)** 3.75 m
- **(E)** 3.2 m
- **(F)** 9.75 m

73. Wax beads are stuck with equal blobs at the same spacing along rods of copper, iron and glass, and one end of each rod is heated equally. On which rod do the beads fall off soonest and on which last? [Heat]

- **(A)** Soonest on copper, last on iron
- **(B)** Soonest on glass, last on iron
- **(C)** Soonest on iron, last on glass
- **(D)** Soonest on glass, last on copper
- **(E)** Soonest on copper, last on glass
- **(F)** Soonest on iron, last on copper

74. A survey of two villages shows: Village A - boys 120, girls 90; Village B - boys 80, girls 110. Which village has more children, and what is the total number surveyed? [Data Handling]

- **(A)** Village A; 380 children in all
- **(B)** Village B; 190 children in all
- **(C)** Village A; 420 children in all
- **(D)** Village A; 210 children in all
- **(E)** Village A; 400 children in all
- **(F)** Village B; 400 children in all
- **(G)** Equal; 400 children in all

75. In a settling tank a scraper works at the bottom and a skimmer at the top. What does each remove? [Wastewater Story]

- **(A)** Both turn sludge into biogas
- **(B)** The scraper colours the water; the skimmer cools it
- **(C)** The scraper adds air; the skimmer adds chlorine
- **(D)** Both remove only sand
- **(E)** The scraper removes floating oil; the skimmer pushes sludge at the bottom
- **(F)** The scraper pushes settled solids (sludge) at the bottom; the skimmer removes floating oil and grease at the top
- **(G)** Both remove germs by heating the water

76. Riya had Rs 200. She spent Rs 56.75 on a book and Rs 89.50 on a toy. How much money is left? [Fractions & Decimals]

- **(A)** Rs 53.50
- **(B)** Rs 53.75
- **(C)** Rs 43.75
- **(D)** Rs 146.25
- **(E)** Rs 54.75
- **(F)** Rs 63.75
- **(G)** Rs 53.25

77. Over four days the temperature rose by 3, 5, 2 and 6 C. On the fifth day it FELL by 4 C. What was the average daily change in temperature over the five days? [Heat]

- **(A)** 2.4 C per day
- **(B)** 2.4 C in total
- **(C)** 3.2 C per day
- **(D)** 1.6 C per day
- **(E)** 20 C per day
- **(F)** 4 C per day
- **(G)** 2 C per day

78. The average age of 4 friends is 12 years. A fifth friend joins and the new average becomes 13 years. How old is the fifth friend? [Data Handling]

- **(A)** 15 years
- **(B)** 25 years
- **(C)** 5 years
- **(D)** 12 years
- **(E)** 17 years
- **(F)** 1 year
- **(G)** 13 years

79. For every 5 litres of sewage treated, a plant produces 2 litres of clean reusable water. How many litres of clean water come from 350 litres of sewage? [Wastewater Story]

- **(A)** 140 litres
- **(B)** 70 litres
- **(C)** 100 litres
- **(D)** 150 litres
- **(E)** 210 litres
- **(F)** 175 litres
- **(G)** 875 litres

80. A tank leaks 0.6 litres every hour. It started with 9 litres. After how many whole hours will less than 3 litres remain for the first time? [Fractions & Decimals]

- **(A)** After 11 hours
- **(B)** After 5 hours
- **(C)** After 15 hours
- **(D)** After 6 hours
- **(E)** After 10 hours
- **(F)** After 12 hours
- **(G)** After 9 hours

81. Five metal samples take 10, 14, 9, 22 and 12 seconds for heat to reach their far end. The shorter the time, the better the conductor. What is the median time, and is the 22-second sample a good or poor conductor compared with the rest? [Heat]

- **(A)** Median 22 s; it is average
- **(B)** Median 12 s; it is the best conductor
- **(C)** Median 12 s; the 22 s sample is the poorest conductor of the five
- **(D)** Median 12 s; it conducts the same as the others
- **(E)** Median 14 s; it is the best conductor
- **(F)** Median 13.4 s; it is the poorest conductor
- **(G)** Median 9 s; it is the poorest conductor

82. In a pictograph one symbol stands for 4 trees. A row has 3 whole symbols and one half symbol. How many trees does the row show? [Data Handling]

- **(A)** 14 trees
- **(B)** 3.5 trees
- **(C)** 13 trees
- **(D)** 16 trees
- **(E)** 7 trees
- **(F)** 12 trees
- **(G)** 10 trees

83. Treated water leaving a plant looks clear. Why is it usually returned to rivers or used for gardens rather than piped straight to taps for drinking? [Wastewater Story]

- **(A)** Treated water turns poisonous inside pipes
- **(B)** It may still carry some germs and chemicals, so it needs further purifying before it is safe to drink
- **(C)** Drinking water must be dirty
- **(D)** It is too pure to drink safely
- **(E)** Rivers make water unsafe, so this is the only option
- **(F)** Garden plants need germs that taps cannot use
- **(G)** Clear water is always safe to drink

84. A medicine dose is 2.5 mL. A bottle holds 0.3 litres. How many full doses are in the bottle? (1 litre = 1000 mL) [Fractions & Decimals]

- **(A)** 150 doses
- **(B)** 0.12 doses
- **(C)** 1200 doses
- **(D)** 120 doses
- **(E)** 12 doses
- **(F)** 30 doses
- **(G)** 75 doses

85. A lab thermometer has its main marks every 1 C, with 5 equal small divisions between them. The liquid stands 3 small divisions above the 26 C mark. What temperature does it show? [Heat]

- **(A)** 26.2 C
- **(B)** 26.6 C
- **(C)** 26.5 C
- **(D)** 26.3 C
- **(E)** 26.06 C
- **(F)** 29 C

86. A shop sold 10 pens at 5 rupees, 10 pens at 8 rupees and 5 pens at 11 rupees. What was the average price per pen sold? [Data Handling]

- **(A)** Rs 8.00
- **(B)** Rs 7.50
- **(C)** Rs 7.00
- **(D)** Rs 6.00
- **(E)** Rs 9.25
- **(F)** Rs 7.40

87. A plant treats 4000 litres. First 10% is lost as sludge, then 20% of what remains is lost in drying. How many litres of water are left after both losses? [Wastewater Story]

- **(A)** 2880 litres
- **(B)** 2520 litres
- **(C)** 1120 litres
- **(D)** 2900 litres
- **(E)** 3600 litres
- **(F)** 2800 litres
- **(G)** 3200 litres

88. A square photo has a side of 0.35 m. A frame strip goes all the way around it. What length of strip is needed? [Fractions & Decimals]

- **(A)** 0.35 m
- **(B)** 1.2 m
- **(C)** 0.7 m
- **(D)** 3.5 m
- **(E)** 0.1225 m
- **(F)** 1.4 m
- **(G)** 1.4 sq m

89. Four identical cans of water are wrapped in white, grey, black and silver foil and left in the sun for the same time. Which order, from greatest to least temperature rise, is correct? [Heat]

- **(A)** Black, white, grey, silver
- **(B)** Black, grey, white, silver
- **(C)** Grey, black, silver, white
- **(D)** White, silver, grey, black
- **(E)** Silver, black, grey, white
- **(F)** Silver, white, grey, black
- **(G)** All rise equally

90. A weather app says the chance of rain tomorrow is 0.35. What is the chance it does NOT rain, written as a fraction in lowest terms? [Data Handling]

- **(A)** 2/3
- **(B)** 3/5
- **(C)** 13/20
- **(D)** 13/100
- **(E)** 17/20
- **(F)** 7/20

91. After digestion, dried sludge is often used as manure. What useful property of this sludge makes it good for soil? [Wastewater Story]

- **(A)** It is full of plastic and metal
- **(B)** It is dried oil and grease only
- **(C)** It contains chlorine gas
- **(D)** It is made only of clean water
- **(E)** It is rich in nutrients that plants need to grow
- **(F)** It poisons all soil it touches
- **(G)** It is pure sand with no nutrients

92. A wire 1.8 m long is cut into pieces each 0.15 m long. How many pieces are there, and is any wire left over? [Fractions & Decimals]

- **(A)** 12 pieces, with none left over
- **(B)** 11 pieces, 0.15 m left over
- **(C)** 9 pieces, 0.45 m left over
- **(D)** 1.2 pieces
- **(E)** 18 pieces, none left over
- **(F)** 12 pieces, with 0.15 m left over

93. A room heater raises a room's temperature by 1.5 C every 4 minutes. The room starts at 16 C and the thermostat switches the heater off at 25 C. For how many minutes does the heater run? [Heat]

- **(A)** 16 minutes
- **(B)** 12 minutes
- **(C)** 60 minutes
- **(D)** 6 minutes
- **(E)** 24 minutes
- **(F)** 13.5 minutes

94. Seven matches had goal totals 2, 0, 3, 2, 5, 2 and 7. Find the mode and the median, then state which is larger. [Data Handling]

- **(A)** Mode 7, median 2 - the mode is larger
- **(B)** Mode 3, median 2 - the mode is larger
- **(C)** Mode 2, median 2.5 - the median is larger
- **(D)** Mode 2 and median 2 - they are equal
- **(E)** Mode 2, median 3 - the median is larger
- **(F)** Mode 2, median 5 - the median is larger
- **(G)** Mode 0, median 2 - the median is larger

95. Two inlet pipes fill a tank: one at 80 litres per minute and one at 120 litres per minute. The tank holds 6000 litres. If both run together, how long to fill it? [Wastewater Story]

- **(A)** 50 minutes
- **(B)** 20 minutes
- **(C)** 75 minutes
- **(D)** 15 minutes
- **(E)** 60 minutes
- **(F)** 30 minutes

96. A shirt marked Rs 800 is sold at 0.85 of its marked price. What is the selling price? [Fractions & Decimals]

- **(A)** Rs 760
- **(B)** Rs 880
- **(C)** Rs 68
- **(D)** Rs 685
- **(E)** Rs 715
- **(F)** Rs 120
- **(G)** Rs 680

97. A spoon at 20 C is placed in soup at 65 C. Which way does heat flow, and when does the flow stop? [Heat]

- **(A)** From the spoon to the soup, until the spoon freezes
- **(B)** From the soup to the spoon, until the spoon is hotter than the soup
- **(C)** Heat flows from the soup to the spoon, and stops when both reach the same temperature
- **(D)** No heat flows, because metal blocks heat
- **(E)** From the spoon to the soup, because the spoon is solid
- **(F)** From the soup to the spoon, for ever

98. The mean of six numbers is 15. If the number 25 is removed, what is the mean of the remaining five numbers? [Data Handling]

- **(A)** 65
- **(B)** 10
- **(C)** 13.5
- **(D)** 15
- **(E)** 13
- **(F)** 12.5
- **(G)** 14

99. In a survey of 250 households, 60% had a flush connected to a covered sewer and the rest used septic tanks. How many households used septic tanks? [Wastewater Story]

- **(A)** 40 households
- **(B)** 100 households
- **(C)** 190 households
- **(D)** 125 households
- **(E)** 60 households
- **(F)** 110 households
- **(G)** 150 households

100. A jug holds up to 1.5 L. You pour in 0.75 L, then pour out 0.4 L, then pour in 0.65 L. How much is in the jug now? [Fractions & Decimals]

- **(A)** 1.05 litres
- **(B)** 1.5 litres
- **(C)** 0.5 litres
- **(D)** 1.0 litres
- **(E)** 0.85 litres
- **(F)** 1.4 litres
- **(G)** 1.8 litres